Lactometer



Category: **Animal Husbandry & Dairy** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Lactometer

What it is used for

To estimate the specific gravity of milk using a lactometer. To obtain a reliable reading by using a clean instrument and measuring milk under the specified temperature conditions.

Main parts

- Lactometer stem and scale
- Weighted bulb
- Milk sample cylinder
- Thermometer
- Clean sample container

Before you start

1. Clean and dry the lactometer and sample cylinder.
2. Collect a representative milk sample in a clean container.
3. Check the lactometer for cracks and a readable scale.
4. Measure the sample temperature and note the method's reference temperature.

How to use

1. Gently mix the milk sample without creating excessive foam.
2. Pour milk into the cylinder without overflowing it.
3. Lower the lactometer gently into the milk so it floats freely.
4. Ensure the instrument does not touch the cylinder sides or bottom.
5. Allow the lactometer to settle and remove any bubbles near the stem.
6. Read the scale at the specified meniscus position and record the temperature.
7. Apply the approved temperature correction if required by the method.
8. Record the result and clean the instrument after use.

⚠ Safety

- Handle the glass instrument carefully to prevent breakage.
- Use clean sampling practices to avoid milk contamination.
- Do not use a cracked or damaged lactometer.
- Follow laboratory procedures for sample disposal.

Care & storage

- Rinse and dry the lactometer after use.
- Store it in a protective case.
- Check scale visibility and instrument condition regularly.
- Keep the cylinder clean and free from residue.

Common mistakes to avoid

- Reading before the lactometer settles.
- Allowing the instrument to touch the cylinder.
- Ignoring sample temperature when correction is required.
- Using a foamy or unrepresentative sample.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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